

Scalable Quantification of White Matter Hyperintensities using Apache Spark and Random Forest

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I. Abstract

White Matter Hyperintensities (WMH) are critical radiological biomarkers for cerebral small vessel disease, stroke, and dementia. As population-level neuroimaging datasets (e.g., UK Biobank) grow to exceed 100,000 subjects, the quantification of WMH burden has transitioned from a clinical task to a significant "Big Data" challenge. Current methodologies face two primary engineering bottlenecks: the "Gold Standard" UBO Detector relies on a legacy, single-threaded MATLAB ecosystem that is difficult to scale, while state-of-the-art Deep Learning approaches require massive datasets and expensive GPU infrastructure.

This project proposes a scalable, cloud-native solution by re-engineering the UBO Detector algorithm into a distributed Apache Spark pipeline. By replacing the legacy k-Nearest Neighbors (k-NN) classifier with a parallelized Random Forest model and utilizing HD-BET for robust skull-stripping, we aim to reduce computational overhead while maintaining diagnostic accuracy. This approach eliminates the dependency on proprietary MATLAB licenses and enables the processing of large-scale MRI datasets on standard CPU clusters, effectively democratizing access to advanced neuroimaging analytics.

II. Introduction

Motivation

As neuroimaging datasets exceed 100,000 subjects, quantifying White Matter Hyperintensities (WMH) [1] has become a "Big Data" challenge. The current "Gold Standard" UBO Detector [2] is ill-equipped for this scale; its legacy, single-threaded MATLAB architecture is computationally inefficient and incompatible with the distributed cloud environments necessary for modern research.

Significance

Although Deep Learning architectures like U-Net offer high accuracy, their reliance on extensive datasets and GPU infrastructure limits scalability. Conversely, ensemble methods like Random Forest [3][4] provide comparable performance

on heterogeneous data with vastly superior computational efficiency. This project addresses the scarcity of scalable neuroimaging tools by transitioning legacy MATLAB workflows to a distributed Spark framework, effectively democratizing access for resource-constrained environments.

Hypothesis

We hypothesize that replacing the UBO Detector with a distributed Apache Spark pipeline will reduce processing time and eliminate proprietary dependencies. We will parallelly evaluate the legacy k-NN classifier against a distributed Random Forest model [5] to ensure comparable segmentation accuracy.

III. Methodology

We propose a distributed pipeline architecture designed to replicate the UBO Detector's logic while resolving its scalability constraints.

Pipeline Architecture

The system is architected as a cloud-native ETL (Extract, Transform, Load) pipeline using Apache Spark.

- **Preprocessing (HD-BET):** Raw T1 and FLAIR MRI scans are first processed using HD-BET (High-Density Brain Extraction Tool) [6]. This neural network-based tool is containerized using Docker to ensure consistent skull-stripping across all nodes, removing non-brain tissue that could confuse the classifier.
- **Feature Extraction:** We replicate the feature space of the original UBO Detector, extracting intensity features (T1/FLAIR ratios) and spatial priors. However, we implement this extraction using PySpark DataFrames, allowing operations to be broadcast across worker nodes rather than looping sequentially.
- **Classification (Random Forest):** We replace the legacy k-NN classifier with a distributed Random Forest model from the Spark MLlib library. Random Forest was selected for its parallel training capabilities and robustness to noise in heterogeneous MRI data.

Tools & Infrastructure

The pipeline is implemented in Python 3.9 using PySpark 3.5. We utilize the University of Utah’s CHPC (Center for High Performance Computing) resources, simulating a cloud environment. The specific cluster configuration consists of 1 Driver Node (32GB RAM) and 4 Worker Nodes (16GB RAM each).

IV. Data

Dataset Selection

We will utilize the **WMH Segmentation Dataset** [7] hosted on Kaggle. This repository provides paired T1-weighted and FLAIR MRI sequences, which are the required inputs for the UBO Detector. Crucially, it includes expert-annotated binary lesion masks that serve as the ground truth for our accuracy evaluation.

V. Evaluation

Success Metrics

We will evaluate the proposed Spark pipeline against the legacy UBO Detector (MATLAB implementation) using three key dimensions:

- 1) **Computational Efficiency:** We will measure the end-to-end processing time per subject. Our success metric is reducing the processing latency compared to the serial MATLAB execution.
- 2) **Scalability:** We will test the pipeline’s throughput (subjects processed per hour) as we scale from 1 to 4 worker nodes. We expect near-linear scalability.
- 3) **Reproduction Accuracy:** Using the UBO Detector’s output as a “Silver Standard,” we will calculate the **Dice Similarity Coefficient (DSC)** between our Spark-generated masks and the MATLAB-generated masks. A $DSC > 0.85$ will indicate successful reproduction of the algorithm.

VI. Task List

We have divided the project into three technical domains: Infrastructure, Data Processing, and Machine Learning.

TABLE I
TASK DISTRIBUTION

Task Description	Owner
Phase 1: Setup	
Lit. Review & Proposal	All Members
Spark Cluster Configuration (CHPC)	Rajavardhan Reddy
Kaggle Data Ingestion Script	Hitesh Kumar
Phase 2: Development	
HD-BET Containerization	Hitesh Kumar
Feature Extraction (PySpark)	Revanth Thoutam
Random Forest Implementation	Rajavardhan Reddy
Phase 3: Evaluation	
Benchmarking (Time/CPU)	Revanth Thoutam
Final Report & Presentation	All Members

VII. Timeline & Deliverables

- **Week 4 (Feb 20):** Project Proposal Finalized. GitHub Repository initialized with skeleton Spark code.
- **Week 6 (Mar 6): Data Pipeline Operational.** Successful ingestion of 5 test subjects (Elderly cohort) and completion of HD-BET skull-stripping.
- **Week 9 (Mar 27): Prototype Check-in.** Feature extraction logic complete. Preliminary Random Forest model trained on the small subset.
- **Week 12 (Apr 17): Full Evaluation.** Pipeline stress-tested on 20+ subjects. Comparison metrics (Time vs. Accuracy) generated.
- **Finals Week:** Final Report, Codebase cleanup, and Demo Presentation.

References

- [1] J. M. Wardlaw *et al.*, “Neuroimaging standards for research into small vessel disease and its contribution to ageing and neurodegeneration,” *The Lancet Neurology*, vol. 12, no. 8, pp. 822–838, 2013.
- [2] J. Jiang *et al.*, “UBO Detector – A cluster-based, fully automated pipeline for extracting white matter hyperintensities,” *NeuroImage*, vol. 174, pp. 539–549, 2018.
- [3] M. Dadar *et al.*, “Validation of T1-weighted based segmentation of white matter hyperintensities in large-scale datasets,” *Human Brain Mapping*, vol. 38, no. 3, pp. 1379–1395, 2017.
- [4] M. F. Rachmadi *et al.*, “Segmentation of white matter hyperintensities using convolutional neural networks with global spatial information,” *NeuroImage: Clinical*, vol. 20, pp. 1165–1177, 2018.
- [5] R. N. Boubela *et al.*, “Big Data Approaches for the Analysis of Large-Scale fMRI Data Using Apache Spark and GPU Processing,” *Frontiers in Neuroinformatics*, vol. 10, 2016.
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- [7] F. Farahmo, “WMH Segmentation Dataset,” Kaggle. [Online]. Available: <https://www.kaggle.com/datasets/farahmo/wmh-dataset>.